

WORKSPACE

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Step 1 of 9 · Data Quality Workflow

# Upload Dataset

Upload the healthcare dataset that will be analyzed, repaired, and evaluated by the framework.



 DATA SOURCE

Uploaded CSV Dataset

 SUPPORTED FORMAT

CSV

 STATUS

Waiting for Upload

Dataset Upload

Select a CSV file to initialize the data-quality assessment workflow.



Click to select a CSV file

Only CSV files are supported.

 icustays.csv

 Uploading Dataset...

# Upload Dataset

Upload the healthcare dataset that will be analyzed, repaired, and evaluated by the framework.



 DATA SOURCE  
Uploaded CSV Dataset

 SUPPORTED FORMAT  
CSV

 STATUS  
Dataset Uploaded

## Dataset Upload

Select a CSV file to initialize the data-quality assessment workflow.



### Dataset Uploaded Successfully

Your dataset is now ready for column selection and quality assessment.

 icustays.csv

UPLOADED DATASET

**icustays.csv**

Successfully loaded and ready for the next layer.

Continue to Column Selection →



# Column Selection

Select a primary column, choose the error taxonomy you want to investigate, and provide related columns when the selected taxonomy requires comparison.

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✓

Upload

2

Column

Assessment

Execution

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DATABASE

icustays.csv

TOTAL COLUMNS

9

REPAIRED COLUMNS

2

✓

Repair Session In Progress

2 columns have already been successfully processed. Select another available column to continue the repair session.

✓ value

✓ medication

Available Columns

Choose the primary column for the quality assessment.

Search columns by name...

All Types

Sort by Name

#

first\_careunit

Categorical

#

hadm\_id

Integer

#

intime

Datetime

#

last\_careunit

Categorical

#

length\_of\_stay

Float

#

outtime

Datetime

#

stay\_id

Integer

#

subject\_id

Integer

#

Unnamed: 0

Integer

Selected Column

Review your selection before starting the assessment.

No column selected

Select an available column from the list to review it before continuing.

Select Error Taxonomy

Choose the type of quality issue you want the system to investigate for the selected column.

Placeholder Values

Detect placeholder values such as N/A or Unknown.

Duplicate Records

Identify duplicate records.

Statistical Anomalies

Unusual numerical values of non medical data.

Implausible Values

Unusual numerical values of medical data.

Temporal Inconsistencies

Conflicting dates and time relationships.

Unit Inconsistencies

Inconsistent measurement units.

Semantic & Contextual Inconsistencies

Unusual textual values of medical data.

Continue to Error Assessment

Available Columns

Choose the primary column for the quality assessment.

9 Columns

Search columns by name...

All Types

Sort by Name

#

**first\_careunit**  
Categorical

#

**hadm\_id**  
Integer

**intime**  
Datetime

#

**last\_careunit**  
Categorical

#

**length\_of\_stay**  
Float

**outtime**  
Datetime

#

**stay\_id**  
Integer

#

**subject\_id**  
Integer

#

**Unnamed: 0**  
Integer

Selected Column

Review your selection before starting the assessment.

#

**length\_of\_stay** ✓  
Float

Select Error Taxonomy

Choose the type of quality issue you want the system to investigate for the selected column.

**Placeholder Values**  
Detect placeholder values such as N/A or Unknown.

**Duplicate Records**  
Identify duplicate records.

**Statistical Anomalies**  
Unusual numerical values of non medical data.

**Implausible Values**  
Unusual numerical values of medical data.

**Temporal Inconsistencies**  
Conflicting dates and time relationships.

**Unit Inconsistencies**  
Inconsistent measurement units.

**Semantic & Contextual Inconsistencies**  
Unusual textual values of medical data.

**Ready for Quality Assessment**  
The system will process the selected taxonomy instead of automatically guessing the error type.

Continue to Error Assessment →



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# Quality Assessment

Assess the column and error taxonomy explicitly selected by the user and present the resulting data-quality evidence and metadata.



## Assessment Configuration

These are the exact selections provided by the user in Column Selection and used as the assessment target.

 DATASET  
**icustays.csv**

 SELECTED COLUMN  
**length\_of\_stay**

 COLUMN TYPE  
**Float**

 SELECTED TAXONOMY  
**Statistical Anomalies**

## Quality Assessment Complete

The selected column and taxonomy have been assessed. The findings and affected records are shown below.

## Assessment Result

Results for **length\_of\_stay** using **Statistical Anomalies**.

Issues Found

### Assessment Findings

#### Statistical Anomalies

Findings identified for the selected taxonomy.

777 Affected Rows

|            |               |               |
|------------|---------------|---------------|
| TOTAL ROWS | AFFECTED ROWS | AFFECTED RATE |
| 10,500     | 777           | 7.40%         |

#### View Affected Records

777 affected records returned by the assessment.



Continue to Execution Mode →

Assessment Result

Results for length\_of\_stay using Statistical Anomalies.

Issues Found

Assessment Findings

Statistical Anomalies

Findings identified for the selected taxonomy.

777 Affected Rows

TOTAL ROWS

10,500

AFFECTED ROWS

777

AFFECTED RATE

7.40%

View Affected Records

777 affected records returned by the assessment.

↓

| ROW | UNNAMED: 0 | SUBJECT_ID | HADM_ID  | STAY_ID  | FIRST_CAREUNIT                                   | LAST_CAREUNIT                                    | INTIME         | OUTTIME        | LENGTH_OF_STAY |
|-----|------------|------------|----------|----------|--------------------------------------------------|--------------------------------------------------|----------------|----------------|----------------|
| 278 | 278        | 10050138   | 30000138 | 40000138 | Medical Intensive Care Unit (MICU)               | Medical/Surgical Intensive Care Unit (MICU/SICU) | 6/25/38 14:40  | 7/11/38 14:01  | 15.97          |
| 290 | 290        | 10050150   | 30000150 | 40000150 | Trauma SICU (TSICU)                              | Trauma SICU (TSICU)                              | 6/22/25 15:07  | 7/8/25 7:39    | 15.69          |
| 312 | 312        | 10050172   | 30000172 | 40000172 | Medical Intensive Care Unit (MICU)               | Medical/Surgical Intensive Care Unit (MICU/SICU) | 9/20/42 15:28  | 10/6/42 14:49  | 15.97          |
| 320 | 320        | 10050180   | 30000180 | 40000180 | Surgical Intensive Care Unit (SICU)              | Surgical Intensive Care Unit (SICU)              | 3/15/11 23:47  | 3/29/11 5:59   | 13.26          |
| 332 | 332        | 10050192   | 30000192 | 40000192 | Surgical Intensive Care Unit (SICU)              | Surgical Intensive Care Unit (SICU)              | 11/25/07 10:25 | 12/8/07 16:37  | 13.26          |
| 338 | 338        | 10050198   | 30000198 | 40000198 | Medical Intensive Care Unit (MICU)               | Medical Intensive Care Unit (MICU)               | 11/11/49 1:22  | 12/1/49 14:03  | 20.53          |
| 343 | 343        | 10050203   | 30000203 | 40000203 | Medical Intensive Care Unit (MICU)               | Medical Intensive Care Unit (MICU)               | 6/18/53 19:01  | 7/9/53 7:42    | 20.53          |
| 346 | 346        | 10050206   | 30000206 | 40000206 | Medical/Surgical Intensive Care Unit (MICU/SICU) | Medical/Surgical Intensive Care Unit (MICU/SICU) | 9/12/47 22:27  | 9/29/47 2:47   | 16.18          |
| 347 | 347        | 10050207   | 30000207 | 40000207 | Medical Intensive Care Unit (MICU)               | Medical/Surgical Intensive Care Unit (MICU/SICU) | 9/10/44 6:59   | 9/26/44 6:20   | 15.97          |
| 360 | 360        | 10050220   | 30000220 | 40000220 | Coronary Care Unit (CCU)                         | Medical Intensive Care Unit (MICU)               | 11/26/38 11:21 | 12/12/38 13:02 | 16.07          |
|     |            |            |          |          |                                                  |                                                  |                |                |                |

Continue to Execution Mode →



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## Repair Execution Mode

Choose how the repair process should be executed for the assessed data.



DATASET

**icustays.csv**

SELECTED COLUMN

**length\_of\_stay**

SELECTED TAXONOMY

**Statistical Anomalies**

AFFECTED ROWS

**777**

### Choose Execution Mode

Select how the repair should be handled after quality assessment.

**Human-in-the-Loop** ☐

Review and approve the repair before it is applied.

- Review each proposed repair
- Approve, modify, or reject
- Changes require user confirmation
- Recommended for medical data review

**Fully Automatic** ☒

Execute the repair automatically without manual approval.

- Executes automatically
- No user interaction
- Fast execution
- Suitable for trusted repair strategies

**Direct Automated Resolution** ☐

Resolve simple, low-impact errors directly before download.

Available only for Placeholder Values and Duplicate Records.

**What Happens Next**

The repair layer will generate and execute repair proposals automatically for the affected records without requiring manual approval for each proposal.

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## Hybrid Repair

Execute the recommended repair strategy and review the repair outcome.



Upload



Column



Assessment



LLM



Execution



Repair



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Dataset

icustays.csv



Selected Column

length\_of\_stay



Detected Issue

Statistical Anomalies



Repair Method

Statistical Repair

### Execute Hybrid Repair

Generate the evidence-based LLM proposals without changing the dataset until the user reviews them.

Generate / Refresh Proposal



### Repair Completed Successfully

The selected repair was validated and applied to the dataset.

### Statistical Anomaly Repair Summary

IQR detected the numerical anomalies. Only the representative before and after values are shown.

777 anomalies

BEFORE VALUE

13.9

AFTER VALUE

2.02

### System-calculated replacement candidates

Calculated from non-anomalous numerical observations.

Fully Automatic

MEAN

3

MEDIAN

2.02

System-selected replacement: Median (2.02)

Repair Method

Statistical Repair

Rows Repaired

777

Confidence Score

90%

Repair Status

Successful



### Continue Repair Process

Review the detailed reasoning in the next layer, or return to the column selection step to repair another column.

Repair Another Column

Continue to Explainability →



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Explainability

Understand how each detected data-quality issue was classified, routed, repaired, and resolved.



| Repair Session Explainability                                                                                        |                |                    |                             |               |            |
|----------------------------------------------------------------------------------------------------------------------|----------------|--------------------|-----------------------------|---------------|------------|
| Each operation below represents one independent column-level repair decision within the same unified repair session. |                |                    |                             |               |            |
| #                                                                                                                    | Column         | Repair Method      | Execution Mode              | Rows Repaired | Status     |
| 1                                                                                                                    | value          | —                  | Direct Automated Resolution | 0             | resolved   |
| 2                                                                                                                    | medication     | —                  | Direct Automated Resolution | 0             | resolved   |
| 3                                                                                                                    | value          | —                  | Direct Automated Resolution | 0             | resolved   |
| 4                                                                                                                    | length_of_stay | Statistical Repair | Fully Automatic             | 777           | Successful |

### Affected Records & Repair Evidence

Every affected record from the complete repair session is explained independently. The recorded repaired value is shown together with its supporting evidence.

**Row-level repair details are not available in the explainability response.**

The operation-level explanation is available above. Once row-level evidence is returned by the repair service, each affected record will appear here independently.

Records  
777

**Repair Outcome**

Successful — 777 rows repaired across the complete repair session.

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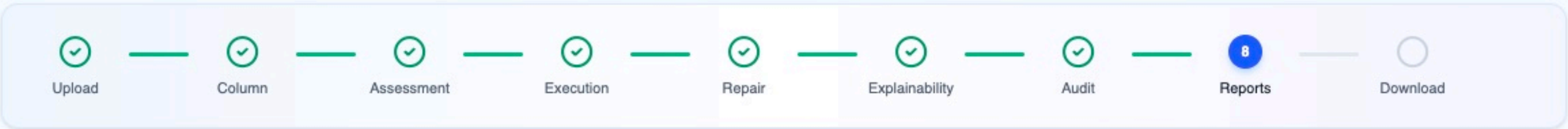
 Reports

LAYER 9 - Download

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# Audit

Chronological trace of every repair operation and its recorded repair evidence.



## Operation Timeline

Chronological record of the repair operations actually executed during this session, with compact statistical-repair evidence where applicable.

1 > **value** Placeholder Values → — resolved

| EXECUTION                   | ROWS REPAIRED | TIME | TIMESTAMP             |
|-----------------------------|---------------|------|-----------------------|
| Direct Automated Resolution | 0             | —    | 9/25/2026, 8:09:56 PM |

2 > **medication** Placeholder Values → — resolved

| EXECUTION                   | ROWS REPAIRED | TIME | TIMESTAMP             |
|-----------------------------|---------------|------|-----------------------|
| Direct Automated Resolution | 0             | —    | 9/25/2026, 8:29:47 PM |

3 > **length\_of\_stay** Statistical Anomalies → Statistical Repair Successful

| EXECUTION       | ROWS REPAIRED | TIME  | TIMESTAMP             |
|-----------------|---------------|-------|-----------------------|
| Fully Automatic | 777           | 0.00s | 9/25/2026, 9:09:27 PM |

3 operations recorded for `lcustays.csv`. Identical backend/session representations of the same repair are consolidated into one audit record. Different column-level repairs remain separate. Statistical Anomalies are shown as a compact before-and-after summary, while other taxonomies retain row-level repair values.

← Back to Explainability

Continue to Reports →



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Reports

Final deliverables generated from the completed EHR data-quality repair session.



Final Repair Summary

Compact overview of the completed repair session.

DATASET  
**icustays.csv**

TOTAL ROWS  
**—**

COLUMNS PROCESSED  
**3**

ROWS REPAIRED  
**777**

SUCCESSFUL OPERATIONS  
**1**

Generated Deliverables

Final files produced by the framework and available for download.

**Repaired Dataset**  
icustays\_repaired.csv

Ready

| FILE TYPE | STATUS    |
|-----------|-----------|
| CSV       | Available |

Contains the final repaired dataset after all selected column-level repair operations have been applied.

- Final records
- All dataset columns preserved
- 3 columns processed
- 777 rows repaired

Download Repaired Dataset

**Audit Log**  
repair\_audit\_log.csv

Ready

| FILE TYPE | STATUS    |
|-----------|-----------|
| CSV       | Available |

Contains the complete operation-level audit history recorded during the repair session.

- 4 repair operations recorded
- Column and detected issue information
- Repair method and execution mode
- Repair status and operation metadata

Download Audit Log

**Final Output Files**

The repaired dataset contains the final repaired data, while the audit log preserves the complete repair-operation history. Both files are available directly from this Reports layer.

← Back to Audit

Finish Workflow